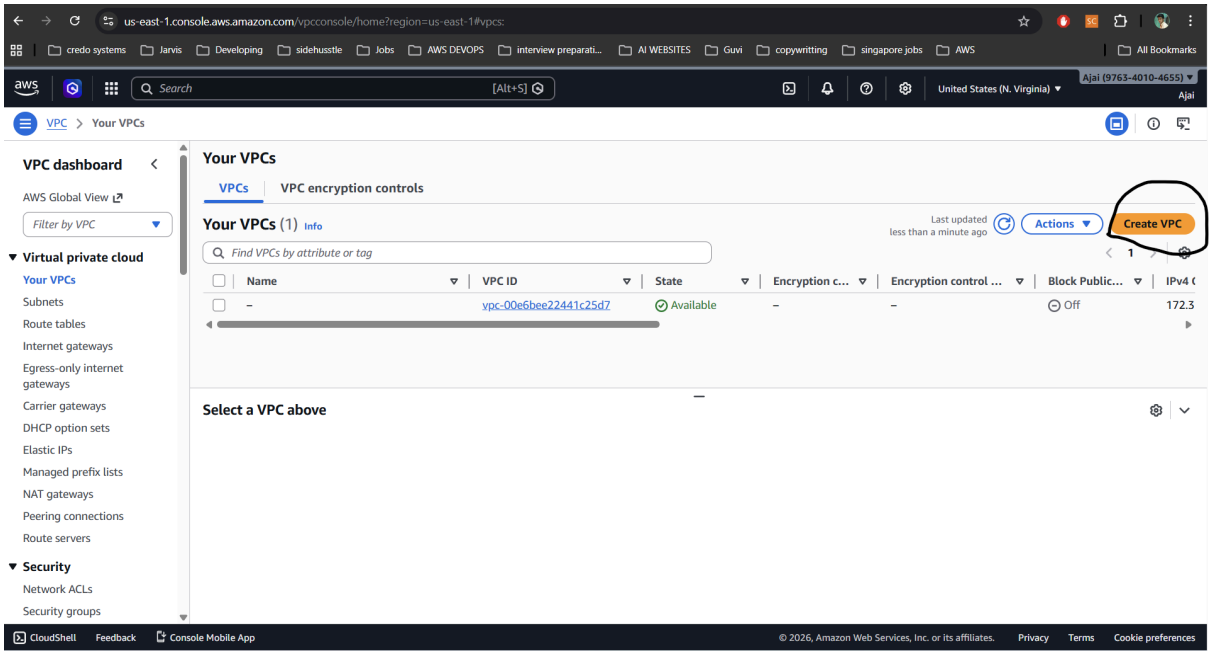
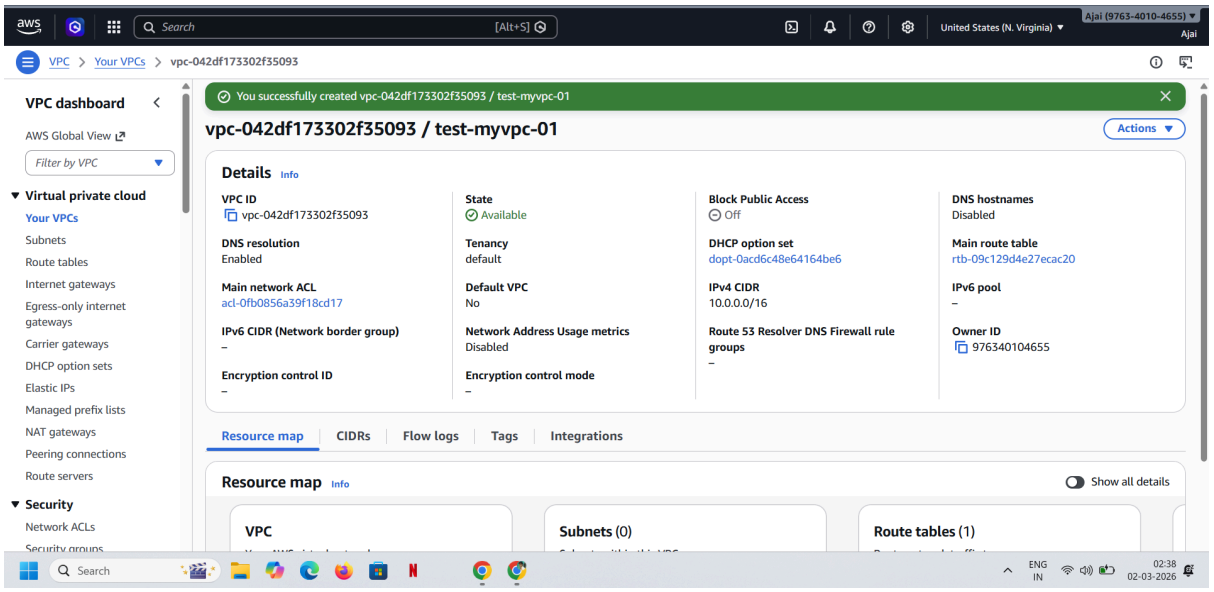
Set up a VPC with an Internet gateway, create a public subnet with 256 IP addresses, a private subnet with 256 IP addresses, make a route table connecting the Internet gateway and the subnets, and launch a Linux EC2 instance by using the above VPC and public subnet.

Set up a VPC with an Internet gateway,

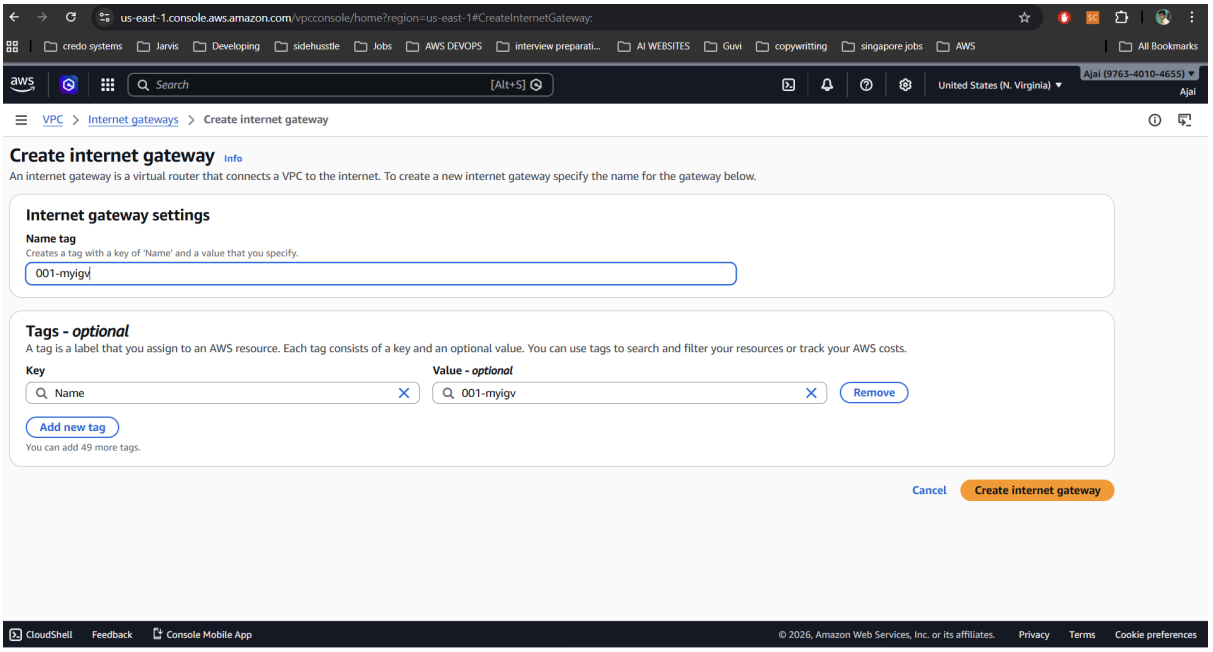


To create a new VPC, navigate to the VPC Dashboard and select **Create VPC**.

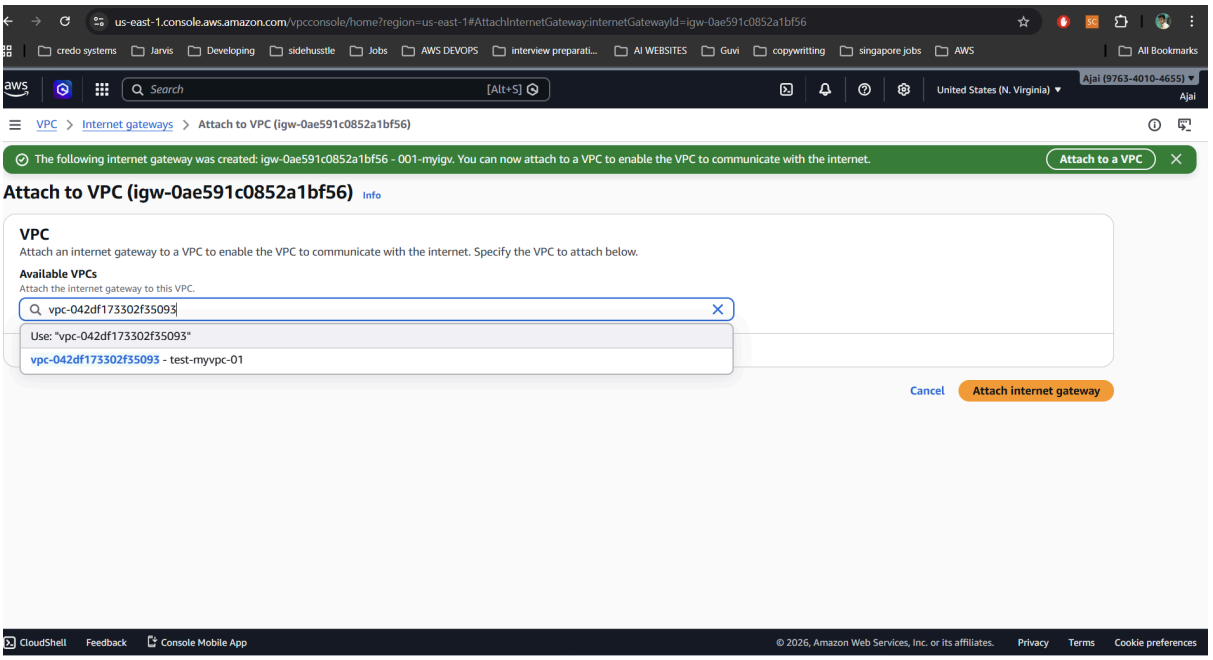


In the configuration settings, assign the name **test-myvpc-01** and specify the IPv4 CIDR block as **10.0.0.0/16**, which provides ample IP address space for your resources. Leave the remaining options at their default values, and then click **Create VPC** to finalize the setup.

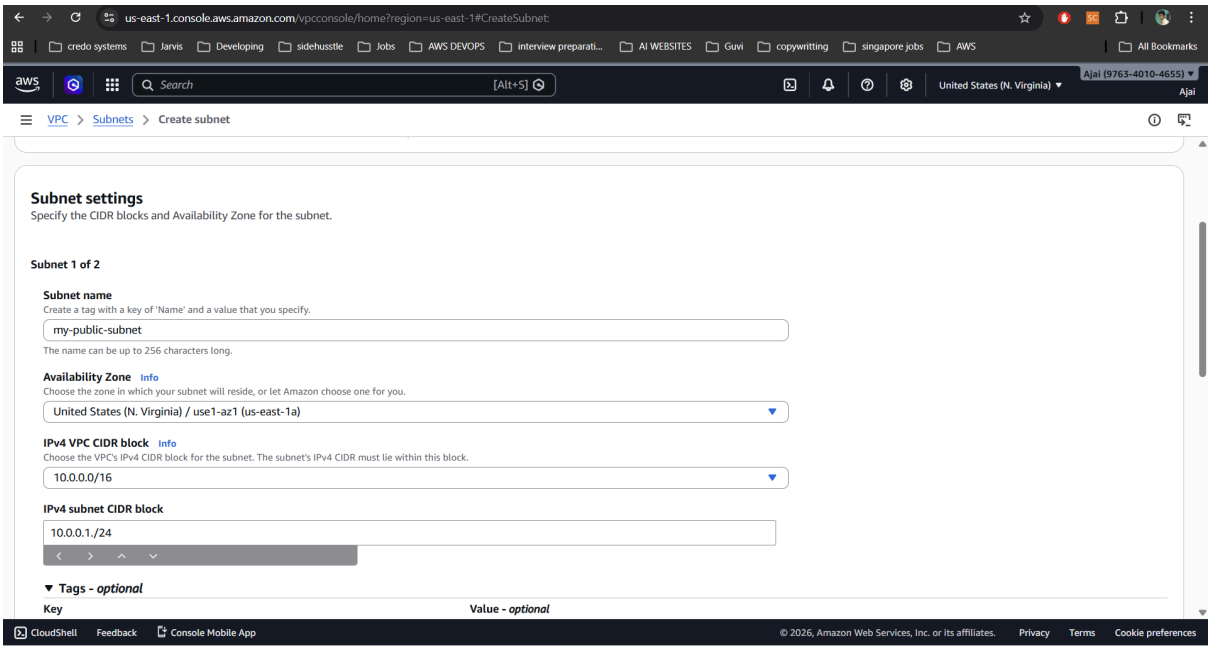
At next I am going to Create and Attach an Internet Gateway



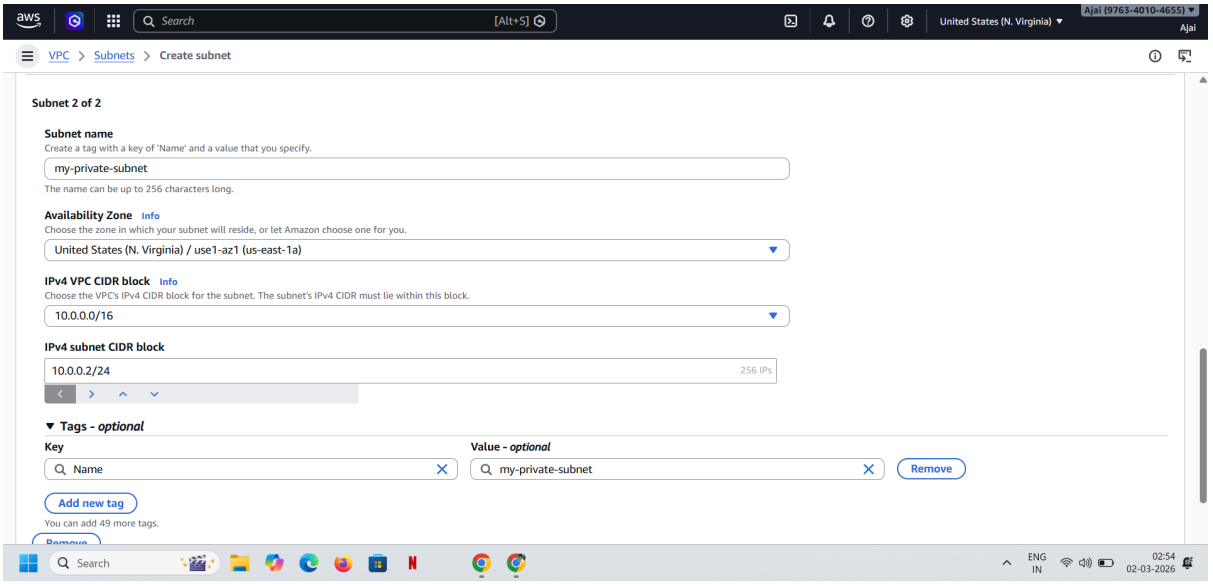
Attached It to my Vpc test-myvpc-01

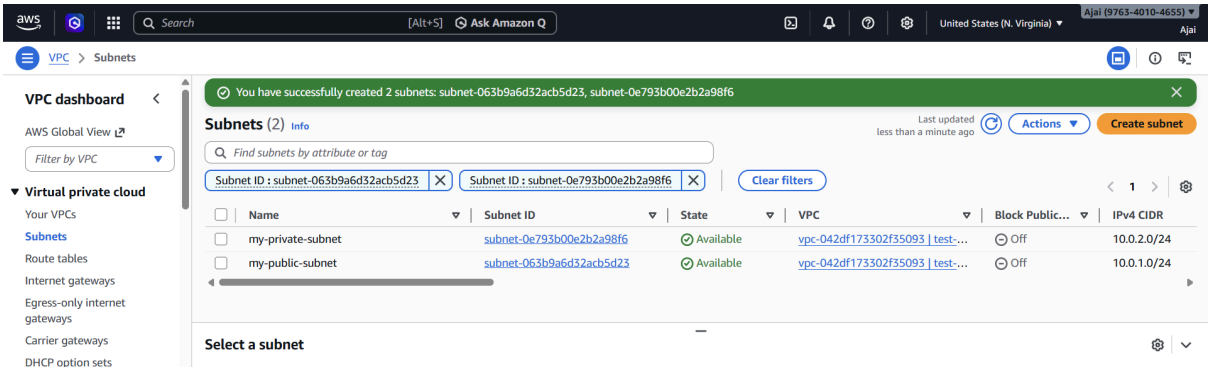


create a public subnet with 256 IP addresses, a private subnet with 256 IP addresses,

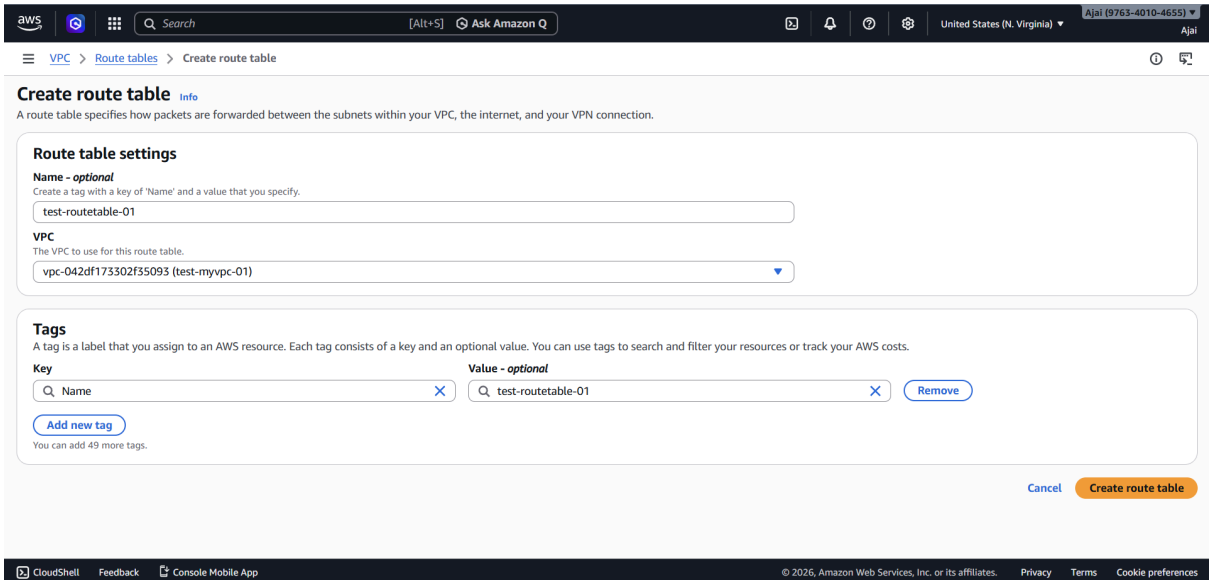


I have created a two subnets of public and private subnets with IP CIDR value of 10.0.1.0.24/ for public and 10.0.2.0/24 for private

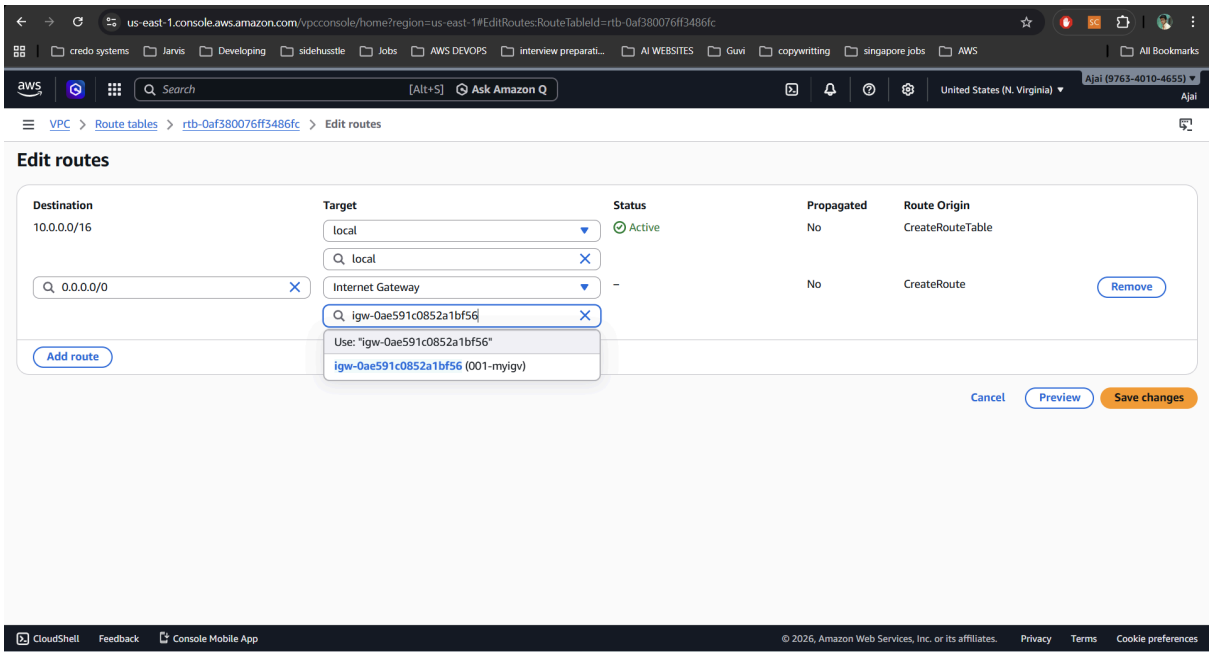




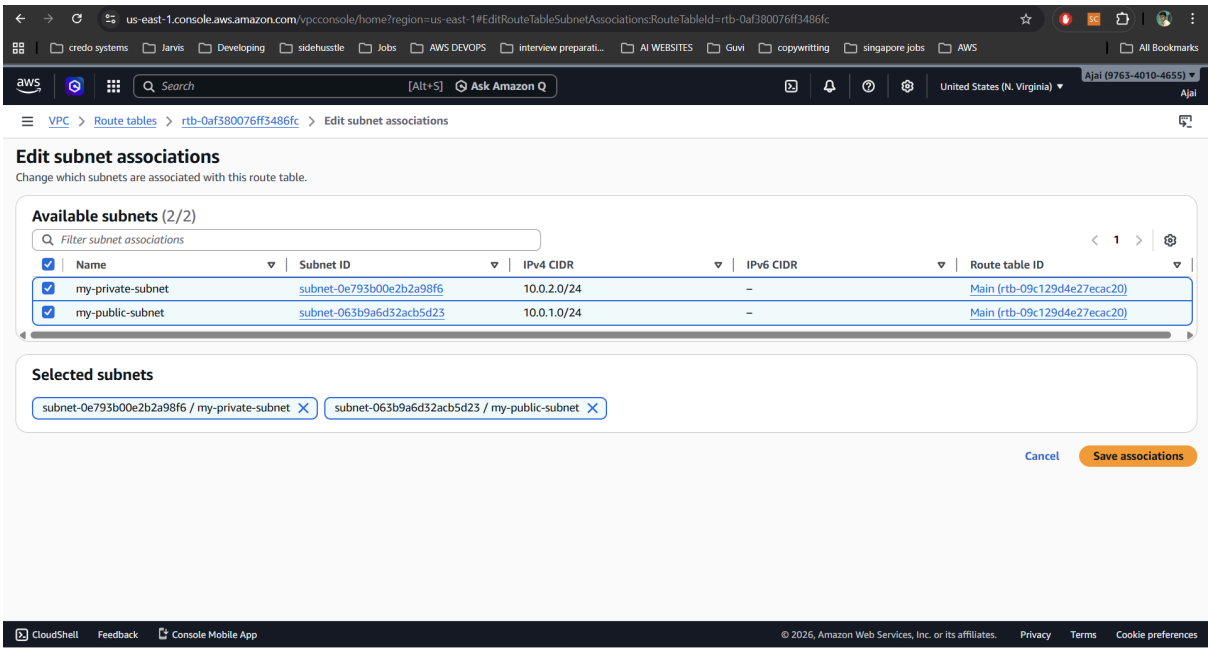
Creating an route table connecting the Internet gateway and the subnets,



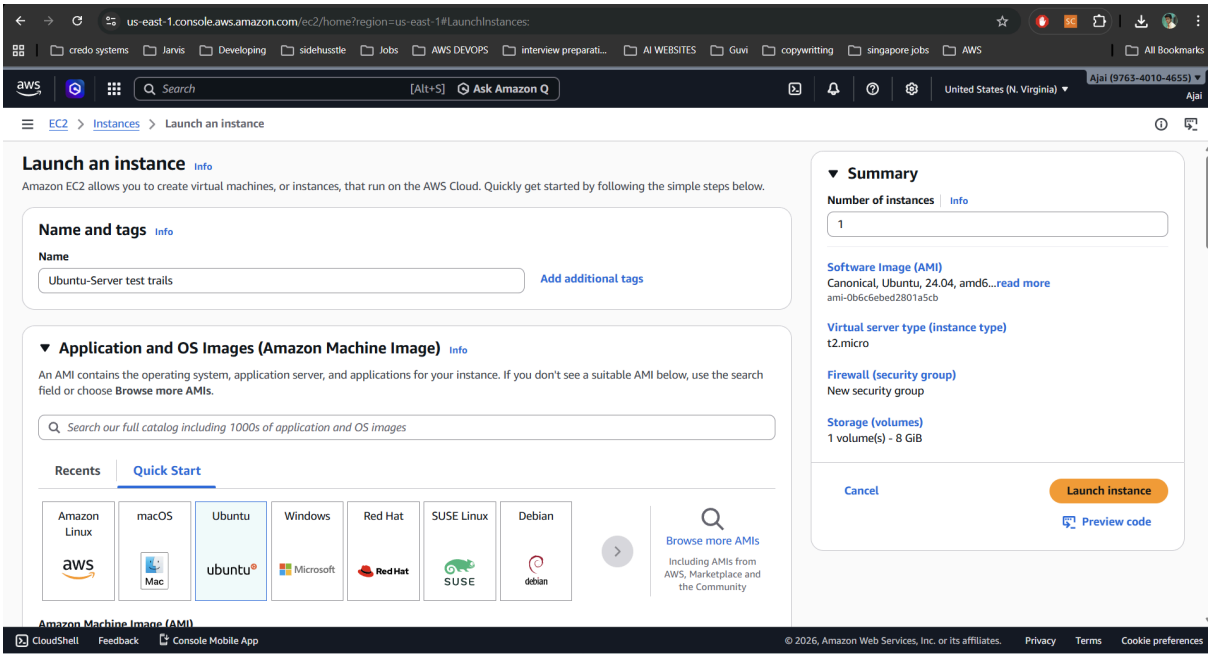
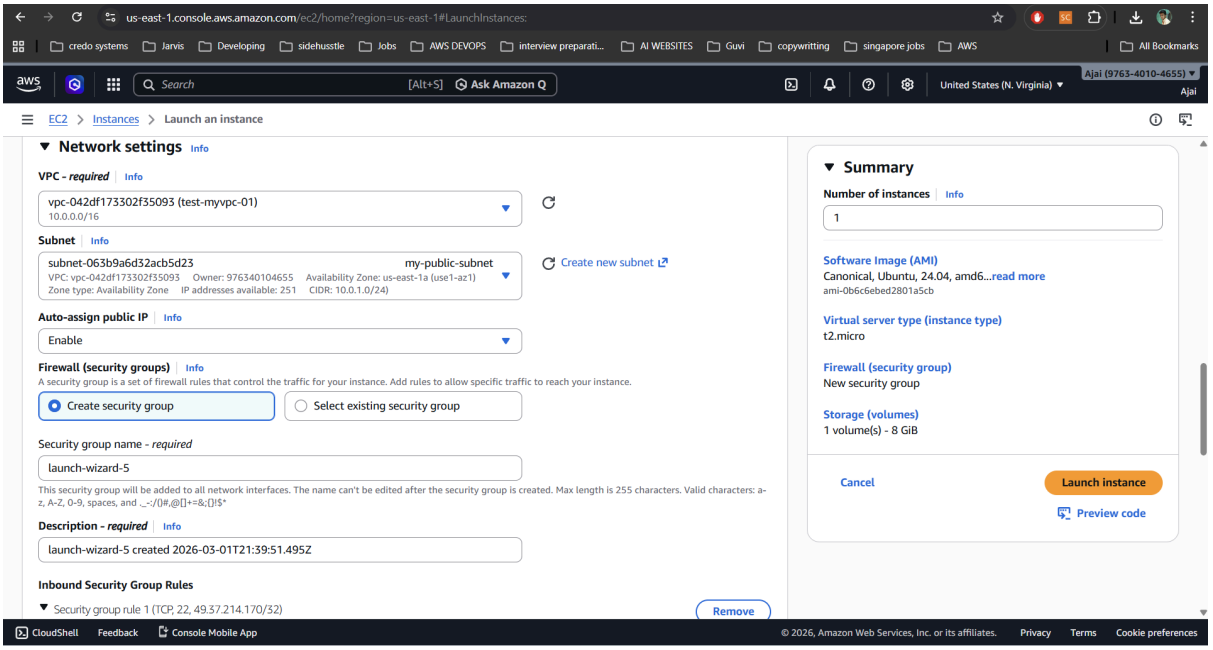
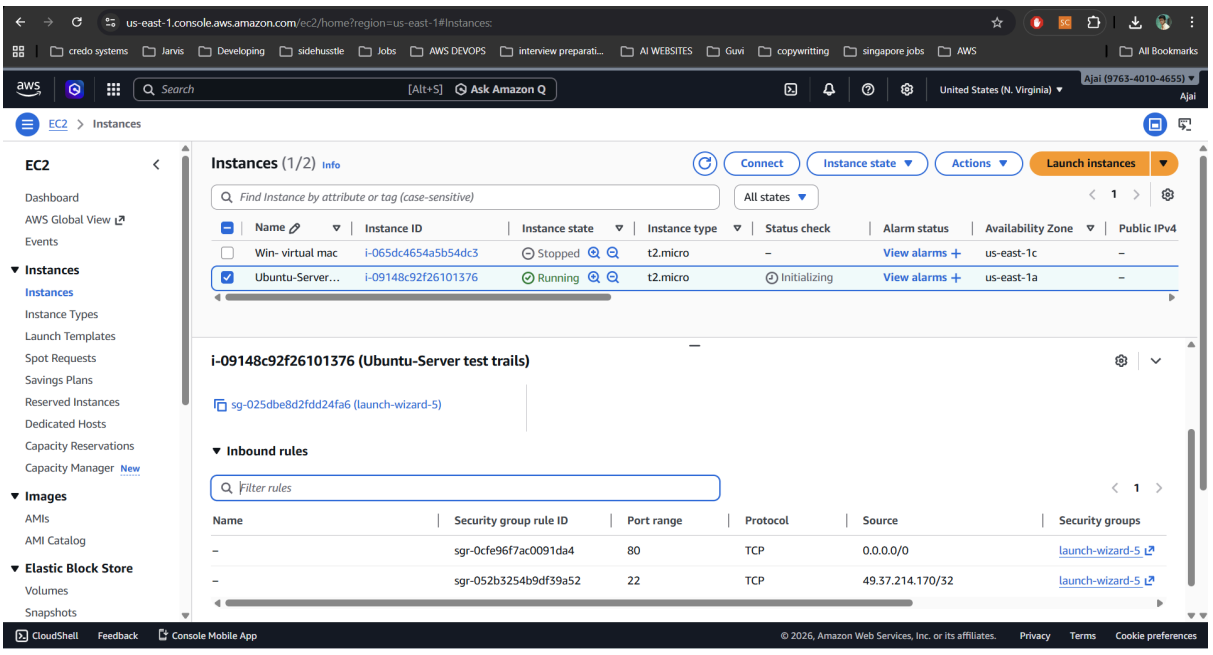
To add an internet route, go to your route table and open the **Routes** tab. From there, click **Edit routes** and choose **Add route**. Set the destination to **0.0.0.0/0** and select 001-**MyIGV** as the target. Once configured, save the routes to establish internet connectivity for your VPC.



Next I will attach my associated subnets into the route table, By navigating a subnet association.



Now I will launch a Linux EC2 instance by using the above VPC and public subnet.



By selecting the VPC and subnets